

Sleep Health and Lifestyle Analysis

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Agenda

Results of sleep health and lifestyle analysis

- Data Description
- Typical Amount of Physical Activity
- Number of Daily Steps
- Distribution of Heart Rates

Data Description

- In the dataset:
 - **`Age` is an example of a continuous variable.**
 - **`Daily Steps` is an example of an integer variable.**
 - **`Blood Pressure` is an example of an ordinal categorical variable.**
 - **`Occupation` is an example of a nominal categorical variable.**

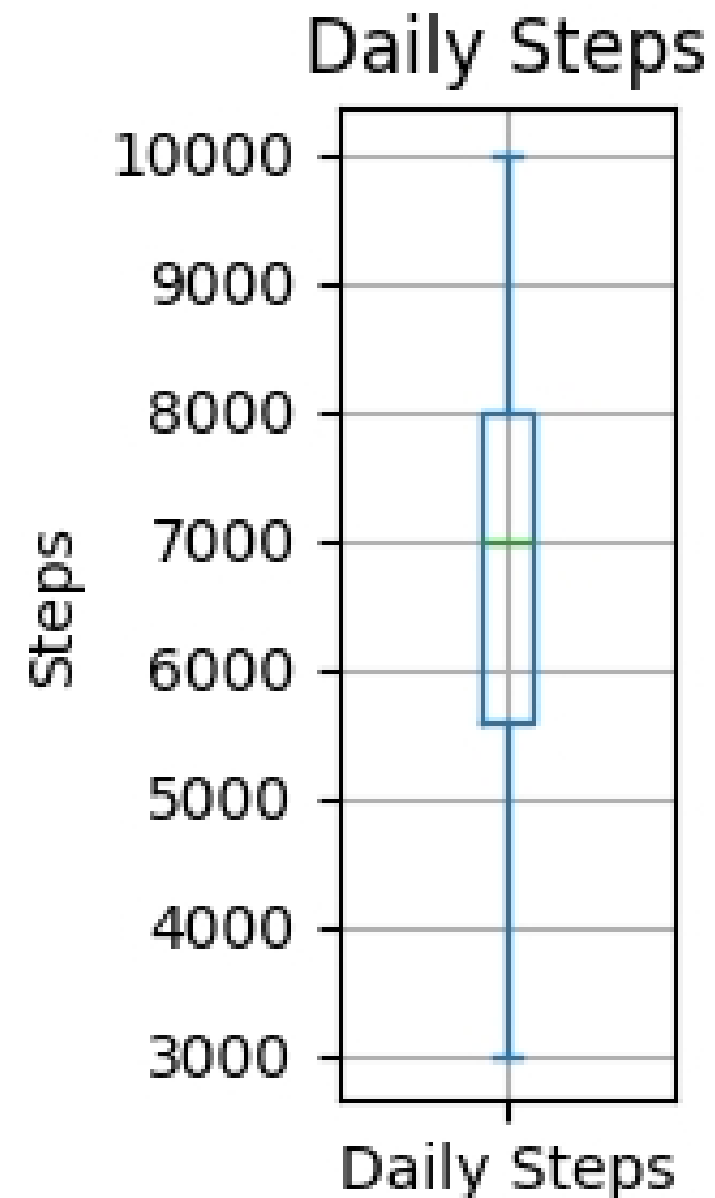
Typical Amount (Minutes) of Physical Activity

	mean	median	mode
Physical Activity Level	59.171123	60.0	60

- The relationships suggest the distribution of physical activity is approximately symmetric and very close to unskewed.

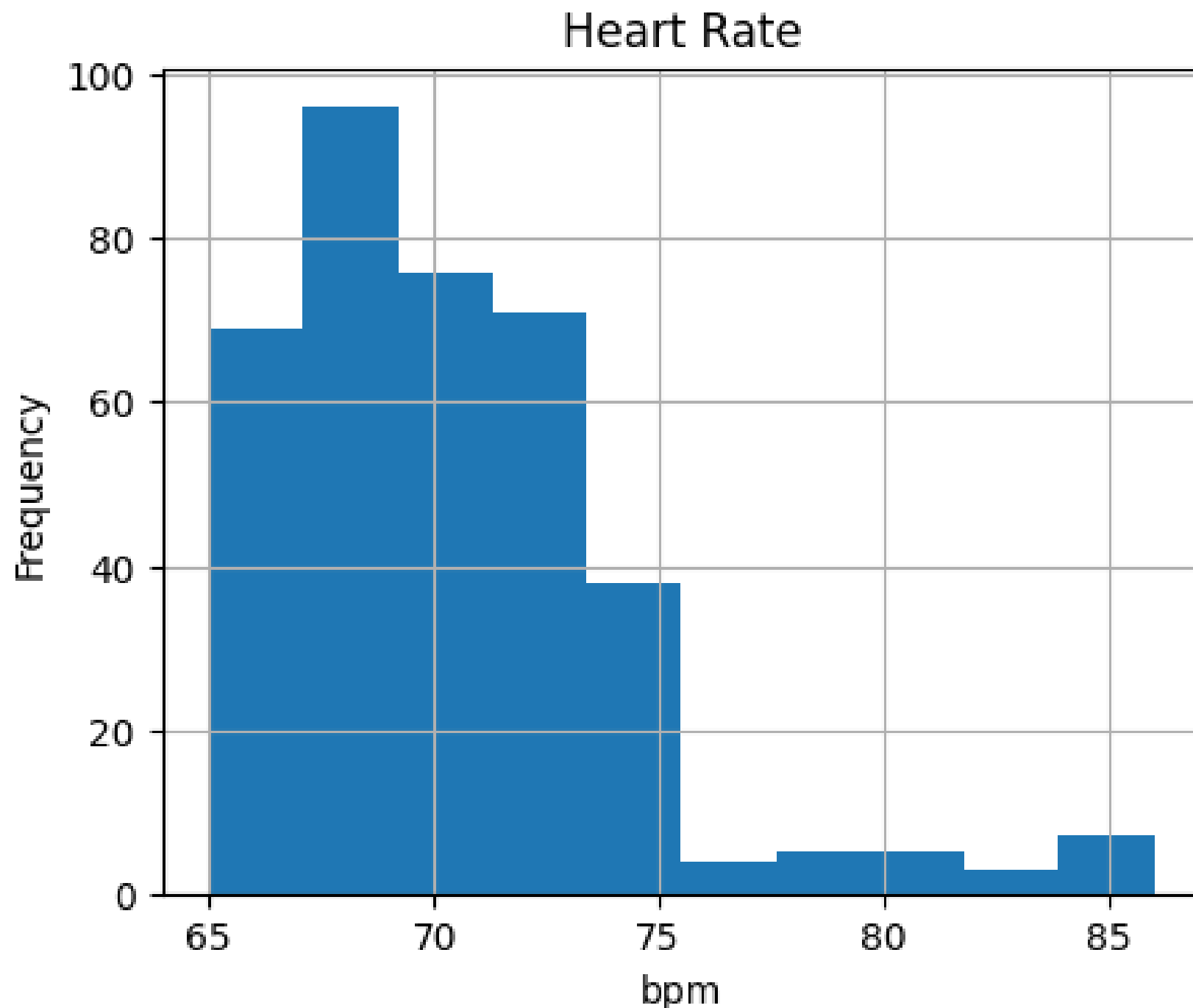
Analysis of Daily Steps Taken

Daily Steps	
count	374.000000
mean	6816.844920
std	1617.915679
min	3000.000000
25%	5600.000000
50%	7000.000000
75%	8000.000000
max	10000.000000



- **Standard deviation: 1617.92 steps**
- **Maximum: 10,000 steps**
- **Minimum: 3,000 steps**
- **Range: $10,000 - 3,000 = 7,000$ steps**
- **Median: 7,000 steps**

Distribution of Heart Rates



- The heart rate distribution is clearly right-skewed (positively skewed), with most observations clustered between about 65 and 73 bpm and a long tail extending to higher values.
- Shape of the distribution
 - The bulk of the bars is on the lower end (mid-60s to low-70s), and the bars gradually thin out as heart rate increases toward the mid-80s, which is characteristic of right skew.
 - There is a single main peak around the high-60s to about 70 bpm, so the distribution is unimodal rather than bimodal or uniform.
- Possible outliers
 - The few bars at very high values (around 80–86 bpm) have very low frequencies compared with the main cluster and can reasonably be considered potential high-end outliers in this context.